

The Geometric Matrix

Rational Derivation of the Pythagorean Triangle

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Abstract

The Pythagorean theorem $a^2 + b^2 = c^2$ describes the relationship between the three sides of a right triangle. This paper adds the constructional context by introducing four variables: the left anchor a , the centroid b , the span c , and the right boundary d . These four variables satisfy a tautological symmetry matrix that defines the triangle configurations generated by a circle under tension. This framework shows that the 3-4-5 triangle is an arithmetic necessity of the specific anchor boundary $[1, 5]$ and demonstrates that any two variables within the system are sufficient to derive the properties of the resulting structure. This construction does not treat the triangle as a free-form object in an unbounded plane; it is a constrained configuration defined over a closed interval $[a, d]$, where its proportions are a necessary consequence of the system rather than an independent selection.

1 The 12-Knot Rope

For over two millennia, the Pythagorean theorem has been the foundation of Euclidean geometry [1]. It states that for a right triangle with legs p and q and hypotenuse r ,

$$p^2 + q^2 = r^2.$$

This relationship is powerful but static; it does not explain why a rope of fixed length pinned at three points must form a specific triangle.

Ancient builders used a 12-knot rope consisting of 12 equal units and 13 knots to create a 3-4-5 right triangle [2][3].

- **System Closure:** The first and last knots are united at the first pin, transforming the rope into a closed circle.
- **Symmetric Spanning:** Under uniform tension, the rope becomes a circle.
- Counting 3 units from the first pin identifies the second pin location, and 4 more units identify the third.
- **Resultant Segment:** The remaining 5 units naturally complete the circle back to the anchor.

The right angle is a physical necessity because the total length (12 units) and chosen spans (3 and 4) leave exactly 5 units to close the circle.

2 The Four Variables

The system defines proportions through a linear anchor boundary $[a, d]$.

- a = Left Anchor
- b = Centroid (arithmetic midpoint)
- c = Span (distance between anchors)
- d = Right Boundary

In the resulting triangle, the three side lengths are b , c , d .

The Symmetry Relations

$$\begin{aligned}a &= b - \frac{c}{2} \\ b &= \frac{a + d}{2} \\ c &= d - a \\ d &= b + \frac{c}{2}\end{aligned}$$

These definitions are not hypotheses; they are the meanings of midpoint and distance.

Given a and d , the values b and c are forced.

Conversely, if we know the centroid b and the span c , we can recover the anchors:

$$a = b - \frac{c}{2}, \quad d = b + \frac{c}{2}$$

Together, these four equations form a closed symmetry matrix. Any one of the four variables can be expressed in terms of the other three. This tautology is the core of the system. It shows that the four points are locked by arithmetic. Any two variables determine the other two.

This closure imposes a structural constraint on the system. The triangle defined by b , c , d is not free-form; it exists within the boundary $[a, d]$, and its proportions are confined to the constraint manifold defined by these relations. Scaling preserves this structure, but does not remove the constraint. The system therefore admits variation in size, but not in form.

2.1 The Delta Constant

The Delta constant (Δ) measures deviation from a right angle.

$$\Delta = b^2 + c^2 - d^2$$

Full Derivation

Substituting $b = \frac{a+d}{2}$ and $c = d - a$:

$$\begin{aligned}\Delta &= \left(\frac{a+d}{2}\right)^2 + (d-a)^2 - d^2 \\ \Delta &= \frac{a^2 + 2ad + d^2}{4} + (d^2 - 2ad + a^2) - d^2\end{aligned}$$

Combine over a common denominator:

$$\begin{aligned}\Delta &= \frac{a^2 + 2ad + d^2}{4} + \frac{4d^2 - 8ad + 4a^2}{4} - \frac{4d^2}{4} \\ \Delta &= \frac{a^2 + 2ad + d^2 + 4d^2 - 8ad + 4a^2 - 4d^2}{4} \\ \Delta &= \frac{5a^2 - 6ad + d^2}{4}\end{aligned}$$

Factor the numerator:

$$5a^2 - 6ad + d^2 = (5a - d)(a - d)$$

Therefore:

$$\Delta = \frac{(5a - d)(a - d)}{4}$$

Since $a < d$, the factor $(a - d)$ is negative.

Classification

- If $d = 5a$ then $\Delta = 0$: Right triangle (Equilibrium)
- If $d < 5a$ then $\Delta < 0$: Obtuse triangle
- If $d > 5a$ then $\Delta > 0$: Acute triangle

2.2 Right Triangles

The condition $\Delta = 0$ with $d = 5a$ defines the right triangle family. For anchor boundary $[a, d] = [n, 5n]$, the sides become:

$$\begin{aligned}b &= \frac{n + 5n}{2} = 3n \\ c &= 5n - n = 4n \\ d &= 5n\end{aligned}$$

This is the 3-4-5 family. Every right triangle generated by this construction is proportionate to 3-4-5.

The 12-knot rope corresponds to the anchor boundary $[1, 5]$, yielding $b = 3$, $c = 4$, $d = 5$.

The right triangle is the motor of the matrix. It is the locked state where the system achieves equilibrium.

The 3-4-5 family emerges from the anchor boundary condition $d = 5a$. Other right triangles, such as the 5-12-13 triangle, have different centroid values and different boundary conditions. They are not examined in this paper.

2.3 Isosceles Triangles

Isosceles triangles occur when two sides are equal. The non-degenerate case occurs when $b = c$. Therefore:

$$\frac{a+d}{2} = d-a$$

Solve:

$$\begin{aligned} a+d &= 2d-2a \\ 3a &= d \end{aligned}$$

Thus the condition is $d = 3a$. Substituting back:

$$\begin{aligned} b &= \frac{a+3a}{2} = 2a \\ c &= 3a-a = 2a \\ d &= 3a \end{aligned}$$

The triangle sides are $b = 2a$, $c = 2a$, $d = 3a$. This is an isosceles triangle with two equal sides of length $2a$ and a base of length $3a$.

For the base case $a = 1$, the sides are 2, 2, 3.

Delta for this family

$$\begin{aligned} \Delta &= (2a)^2 + (2a)^2 - (3a)^2 \\ \Delta &= 4a^2 + 4a^2 - 9a^2 \\ \Delta &= -a^2 \end{aligned}$$

Since $\Delta < 0$, this family is inherently obtuse.

2.4 Equilateral Triangles

To evaluate the Delta relation for an equilateral triangle, let all three side lengths be a constant s .

Here we are evaluating the Delta relation itself, not mapping the equilateral triangle from an anchor boundary $[a, d]$.

1. Identity of Squares

In an equilateral triangle, all sides are a constant s . Therefore, for the Delta evaluation, all squares are identical:

$$b^2 = s^2, \quad c^2 = s^2, \quad d^2 = s^2$$

2. The Delta (Δ) of the Equilateral

Applying the Matrix relation $b^2 + c^2 - d^2 = \Delta$, the result is always the value of a single square:

$$s^2 + s^2 - s^2 = s^2$$

Therefore:

$$\Delta = s^2$$

Unlike our [2:2:3] family, where Delta was negative ($-a^2$), in the equilateral triangle Delta is positive and equal to the square of the side. This places it firmly in the acute domain, far from the Pythagorean boundary.

3. The Squared Height (h^2)

The derivation of the height in the equilateral triangle does not depend on decimal approximations, but rather on the fraction of its squares:

$$\begin{aligned} h^2 &= s^2 - \left(\frac{s}{2}\right)^2 \\ h^2 &= s^2 - \frac{s^2}{4} \\ h^2 &= \frac{3}{4}s^2 = 0.75s^2 \end{aligned}$$

4. Structural Interpretation

The Delta evaluation shows that the equilateral configuration produces the maximum positive value among the symmetric comparison cases examined here. While the right triangle is the equilibrium point ($\Delta = 0$), the equilateral triangle is the fully symmetric case in which the three side contributions are identical, resulting in $\Delta = s^2$.

This section evaluates the Delta diagnostic for the equilateral geometry; it does not assert that a non-degenerate equilateral triangle is generated by the anchor-boundary mapping defined in Section 2.

3 The Circumcenter

We locate the circumcenter S via rational partitioning. No trigonometry is required, and no Heron's formula is needed.

Place the triangle with base c on the x-axis:

$$V_1 = (0, 0), \quad V_2 = (c, 0), \quad V_3 = (x, y)$$

The distance from V_1 to V_3 is b :

$$x^2 + y^2 = b^2 \tag{1}$$

The distance from V_2 to V_3 is d :

$$(x - c)^2 + y^2 = d^2 \tag{2}$$

Subtract (1) from (2):

$$(x - c)^2 - x^2 = d^2 - b^2$$

Expand:

$$(x - c)^2 = x^2 - 2cx + c^2$$

Therefore:

$$\begin{aligned}(x^2 - 2cx + c^2) - x^2 &= d^2 - b^2 \\ -2cx + c^2 &= d^2 - b^2\end{aligned}$$

Solve for x :

$$x = \frac{b^2 + c^2 - d^2}{2c}$$

From equation (1):

$$y = \sqrt{b^2 - x^2}$$

The perpendicular bisector of the base V_1V_2 is the vertical line at:

$$X_c = \frac{c}{2}$$

The circumcenter S lies at the intersection of this line and the perpendicular bisector of V_1V_3 .

Compute the midpoint of V_1 and V_3 :

$$M_{13} = \left(\frac{x}{2}, \frac{y}{2}\right)$$

The slope of V_1V_3 is y/x , so the slope of the perpendicular bisector is $-x/y$.

The equation of the perpendicular bisector of V_1V_3 is:

$$Y - \frac{y}{2} = -\frac{x}{y} \left(X - \frac{x}{2}\right)$$

Substitute $X = c/2$:

$$\begin{aligned}Y_c - \frac{y}{2} &= -\frac{x}{y} \left(\frac{c}{2} - \frac{x}{2}\right) \\ Y_c - \frac{y}{2} &= -\frac{x}{y} \times \frac{c - x}{2}\end{aligned}$$

Therefore:

$$Y_c = \frac{y}{2} - \frac{x(c - x)}{2y}$$

The circumradius R is the distance from $S\left(\frac{c}{2}, Y_c\right)$ to $V_1(0, 0)$:

$$R = \sqrt{\left(\frac{c}{2}\right)^2 + (Y_c)^2}$$

All quantities are derived from the variables b , c , d through rational operations and one square root.

4 The Origami Principle

The circle is a folded state of a square.

Given a square of side length D , the circular circumference C is defined by the rational harmonic fraction:

$$C = \frac{22}{7}D$$

Since the diameter $D = 2R$, the circumference is:

$$C = \frac{44}{7}R$$

The bounding square is recovered via $D = 2R$. The fraction $22/7$ is a rational constant that arises from discrete folding geometry; it is not a claim about the value of π in continuous Euclidean geometry, nor does it assert that $\pi = 22/7$.

A full derivation is published in Johnson (2026) [5].

5 The Bounding Square

Standard geometry treats triangles, circles, and squares as separate primitives. The system shows they are states of one system:

$$\begin{aligned} \text{Variables} &\rightarrow \text{Triangle} \\ \text{Triangle} &\rightarrow \text{Circumcircle} \\ \text{Circumcircle} &\rightarrow \text{Bounding Square } (D = 2R) \end{aligned}$$

The square is the primitive state. The circumcircle is the circle that passes through the triangle's three vertices. The triangle is defined by three points pinned to that circumcircle.

In standard Euclidean geometry, specifying two sides of a triangle leaves the third side undetermined; the hinge remains open. One needs an additional piece of information such as an angle, a height, or a third side. This is accurate for triangles existing in an open void.

However, in the present system, the triangle exists within a bounding square, and this boundary provides the missing information. Structural determinacy is a result of the boundary it occupies. The square defines the limit, the circumcircle defines the path, and the triangle is the resulting pinned state.

6 Scaling Law

For the right triangle family where $d = 5a$, let $a = n$. The proportions scale linearly:

$$a = n, \quad b = 3n, \quad c = 4n, \quad d = 5n$$

This means that if one has a rope of length $12n$ units, one can construct a similar right triangle with sides scaled by n .

For the isosceles family where $d = 3a$, scaling $a = n$ gives sides $2n, 2n, 3n$.

For any triangle generated by the system, scaling the anchor boundary $[a, d]$ by a factor k scales b and c by the same factor, thereby preserving the shape.

This property allows modular assembly:

- Horizontal extension by sharing anchor boundaries $[a, d]$
- Vertical stacking by aligning centroids b

The scaling law is trivial in algebra but essential in construction.

7 Honoring Pythagoras

Pythagoras of Samos and his followers are credited with the first general proof of the theorem $b^2 + c^2 = d^2$ [1][4]. That theorem is a landmark of human reasoning, and the present work does not diminish it.

Rather, this work places the theorem in a larger context of a circle under tension. The Pythagorean equation remains the condition for a right angle, $\Delta = 0$, while the present system adds the boundary variables $a = b - c/2$ and $d = b + c/2$. This shows that right triangles proportionate to 3-4-5 are the result of pinning a circle within the specific anchor-boundary construction examined here.

Pythagoras gave us the relationship; this work adds the boundary variables that place that relationship within the construction.

The Pythagorean theorem remains intact. The Geometric Matrix adds constructional context around the relationship.

8 Conclusion

The Geometric Matrix adds the boundary variables a and d to the three-side relationship expressed by the Pythagorean theorem [1][5]. It shows that the 3-4-5 triangle is an arithmetic necessity of the boundary $[1, 5]$ [2][3].

The Delta constant measures structural drift, classifying triangles as acute, right, or obtuse without trigonometry. The circumcenter is located through rational partitioning and works for every triangle generated by the system. The circle is treated as the folded state of a square, yielding the rational circumference $C = \frac{22}{7}D$, with full derivation in Johnson (2026) [5].

The bounding square closes the hinge, providing determinacy, and the scaling law enables modular construction.

While Euclidean geometry serves the void, the system provides the boundary that ensures structural determinacy, restoring the builder's perspective to the mathematician's abstraction.

References

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- [2] Petrie, W.M.F. *The Pyramids and Temples of Gizeh*. Field & Tuer, London, 1883.
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